

Robotic Arm Quality Inspection

4-DOF arm IK + conveyor cell,
classical OpenCV and ML defect
detection, auto reports

AT A GLANCE

96.0 %
Classical acc (150 test)

97.3 / 90.0 %
SVM / RF acc (150 test)

0 %
False-alarm rate (all 3)

24/24
Cell: pass/fail correct

21.5/min · 1.59 s
Throughput / cycle

10 um
Numeric IK max error

15/15
Unit tests

VERDICT

SIM PASS **TESTS 15/15**

IK VERIFIED

DECLARED SUBSTITUTION

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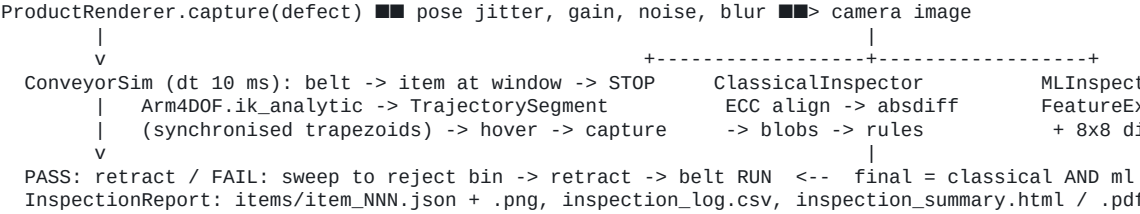
1. Objective and brief mapping

I built a simulated quality-inspection cell: a conveyor of synthetic PCB-like tiles with five injected defect types, a 4-DOF articulated arm (base yaw, shoulder, elbow, wrist pitch) that positions a downward-looking camera over each stopped item, two independent defect detectors (a rule-based OpenCV pipeline and scikit-learn classifiers on HOG + colour + diff features) and an automated per-item report (JSON, CSV, HTML and PDF with annotated crops). The conveyor loop measures cycle time, throughput and arm utilisation.

Every number here was produced in this container by `sim/run_sim.py` and the unittest suite (`sim/metrics.json`, `sim/inspection_report/`, `sim/test_output.log`). The ML evaluation uses a disjoint train set (240 images) and test set (150 images) generated from different seeds.

2. Architecture

`src/arm_inspect/` is imported unchanged by `sim/run_sim.py`, the tests, the ROS 2 node and the RoboDK / CoppeliaSim scripts.



3. Implementation details

- **Products:** 128 px tile with 3 x 3 copper pads, one plated hole and a silkscreen label. Nuisance per capture: ± 3 px shift, $\pm 3^\circ$ rotation, 0.85-1.15 gain plus a lateral gradient, $N(0, 4)$ sensor noise, optional 3 x 3 blur. Defects: bright/dark scratch line, undrilled hole, substrate tint shift, corner chip (14-26 px), label shifted / wrong / truncated.
- **Classical pipeline:** `cv2.findTransformECC` (Euclidean, 5-level pyramid) registers the capture to the golden tile; gain-normalised per-pixel max-channel absdiff; threshold 58 (the ECC residual around the high-contrast hole ring peaks near 45, real defects exceed 68); 3 x 3 closing (no opening, it erased 1 px scratches); connected components with area, elongation, hole overlap, label overlap, corner/edge contact. Rules in order: chroma shift > 0.045 -> colour_dev; hole overlap > 0.45 -> missing_hole; compact corner blob -> chipped_edge; elongation > 3 -> scratch; label NCC < 0.78 or label-ROI blob area > 30 -> misprint.
- **ML:** skimage HOG (16 px cells, 2 x 2 blocks) + 12 x 6 HSV histogram over the board + 8 x 8 down-sampled diff map + substrate chromaticity = 1903 features; RandomForest (200 trees) and an RBF-SVM (C = 5, standardised inputs). Fit time 5.1 s / 7.4 s; prediction 21 / 28 ms per item including alignment.
- **Arm:** classic DH table ($d_1 = 0.30$, $L_2 = 0.35$, $L_3 = 0.30$, $L_4 = 0.10$ m), FK by chained transforms, analytic IK (base yaw by atan2, wrist centre from the tool pitch, law of cosines for the 2-link, elbow-up with elbow-down fallback, joint-limit check) and a damped-least-squares numeric IK on the 4 x 4 numeric Jacobian [x, y, z, pitch] with restarts from canonical seeds when a joint limit stalls the iteration.
- **Trajectories:** per-joint time-optimal trapezoid (or triangle) profiles, stretched to the slowest joint so all joints start and stop together; velocity limits 180/120/150/240 deg/s. Cycle: approach 0.4-0.6 s, 150 ms exposure + processing, optional reject sweep, retract.

4. Verification

`python3 -m unittest discover -s tests -v: 15 tests, all passing.`

Test group	What is checked	Result
Kinematics (7)	FK at zero / shoulder-up / yawed poses, analytic IK round trip on 150+ random targets (1e-9 m, limits respected), numeric IK converges to 1e-4 m from a bad seed, unreachable target -> None, clamp	pass
Trajectory (2)	trapezoid vs triangle profile algebra, segment end-points exact, all joints within velocity limits and synchronised	pass
Inspection (5)	ECC alignment cuts the residual below 25 %, golden passes, each defect type detected on 4/5 clean injections, classical accuracy ≥ 85 % with false-alarm rate ≤ 10 %, RF train/test split ≥ 80 %	pass
Conveyor (1)	6-item loop: all inspected, throughput > 5/min, pass/fail ≥ 80 %, joint limits respected on every log sample	pass

5. Detection results on the 150-image test set (25 per class, seeds disjoint from training)

Overall accuracy: classical 96.0 %, RF 90.0 %, SVM 97.3 %. Binary defect detection (any defect vs ok): classical recall 98.4 %, RF 90.4 %, SVM 96.8 %, all three with 0 false alarms on the 25 good tiles. The weak class is **scratch**: thin 1 px lines whose contrast falls under the diff threshold after blur are missed by the rules (2/25) and confused with misprint when they cross the label (4/25); the RF misses 10/25

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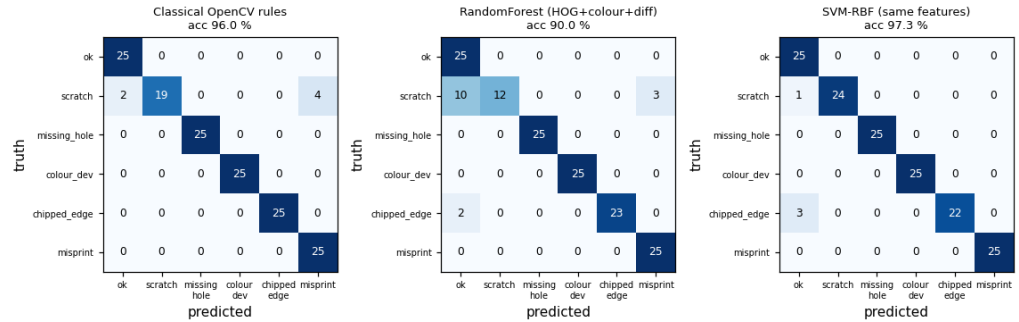
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scratches because HOG at 16 px cells barely sees a 1 px line, while the SVM on the same features recovers 24/25. The misprint column being perfect is partly a property of the synthetic data (the label ROI is known exactly); on a real line I would expect this to drop first.

Class	Classical P / R (%)	RF P / R (%)	SVM P / R (%)
ok	93 / 100	68 / 100	86 / 100
scratch	100 / 76	100 / 48	100 / 96
missing_hole	100 / 100	100 / 100	100 / 100
colour_dev	100 / 100	100 / 100	100 / 100
chipped_edge	100 / 100	100 / 92	100 / 88
misprint	86 / 100	89 / 100	100 / 100



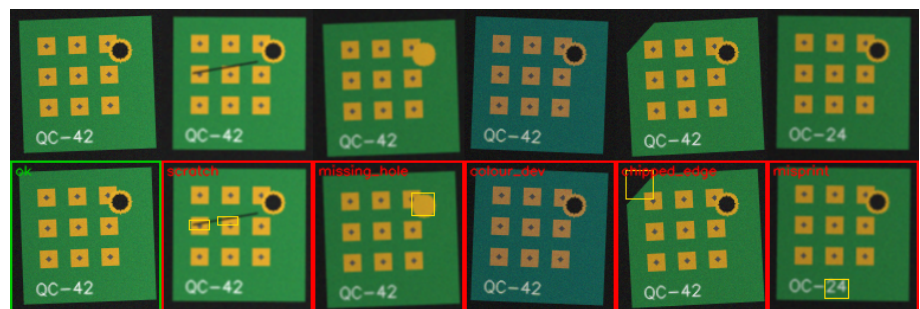
Confusion matrices (rows = truth, columns = prediction) of the three detectors on the same 150 test images.

6. Cell performance (24-item conveyor run, seed 3)

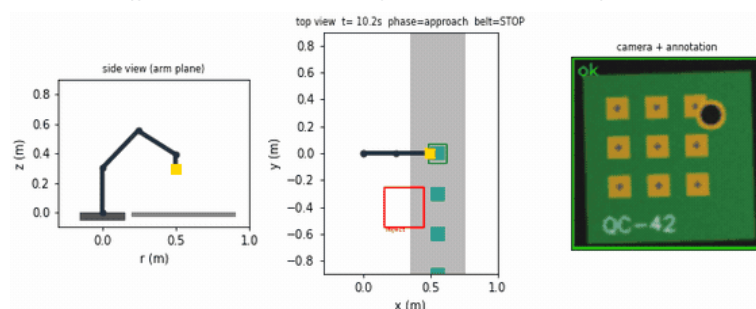
The belt stops for every item, so throughput is bounded by the arm cycle (approach + exposure + optional reject sweep + retract), not by the 0.25 m/s belt. The final verdict is the AND of both detectors: an item passes only if the rules and the RandomForest both call it good, which trades a little recall on the weak scratch class for the zero-false-alarm behaviour a rejection station needs.

Metric	Value
Items inspected	24 / 24
Simulated run time	66.85 s
Throughput	21.54 items/min
Cycle time mean / min / max	1.589 / 1.25 / 1.9 s
Arm utilisation (busy / total)	55.7 %
Rejected / passed	12 / 12
Pass/fail verdict correct	24 / 24
Defect type named correctly (classical)	23 / 24
Analytic IK max position error (250 targets)	2.3e-16 m
Numeric IK max error / mean iterations	9.9e-06 m / 6.5

7. Figures



Top row: one capture per class (ok, scratch, missing hole, colour deviation, chipped edge, misprint). Bottom row: the classical inspector's annotation (green = PASS, red = FAIL, yellow boxes = diff blobs).



GIF frame: side view of the arm in its plane, top view with the belt (grey), inspection window (green), reject bin (red) and queued tiles, and the camera view with the verdict.

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02

Robotics & Automation
Codec Technologies India
Internship 2026

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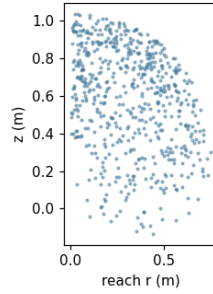
SIM PASS

TESTS 15/15

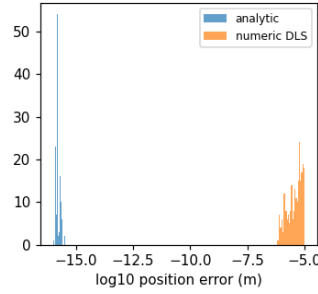
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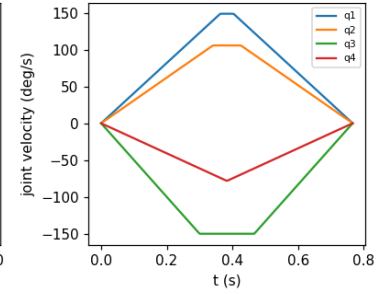
reachable (r, z) samples, random q



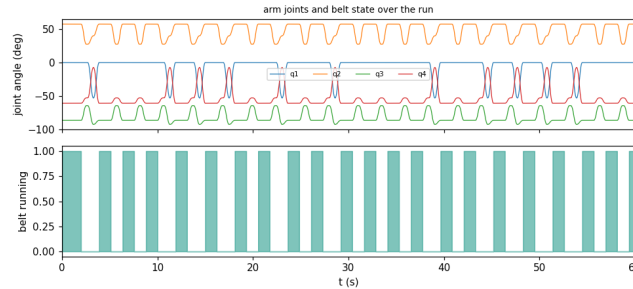
IK error, 250 targets (numeric: 6.5 it avg)



synchronised trapezoids, T = 0.77 s



Left: reachable (r, z) samples from random joint vectors. Middle: log10 position error of analytic vs numeric IK on 250 targets. Right: synchronised trapezoidal joint velocities for one move.



First 60 s of the run: the four joint angles (top) and the belt run/stop state (bottom). Every stop is one inspection cycle; longer stops include the reject sweep.

7b. Per-item log excerpt (sim/inspection_report/inspection_log.csv, first 10 of 24)

The truth column is hidden from both inspectors and is only joined in afterwards for scoring. The alignment column shows the ECC pose estimate that the diff is computed under; the injected jitter is up to 3 px and 3 degrees. Item 0 is a good example of why both detectors are used: the rules caught the scratch with confidence 1.0 while the RF called it ok at 0.37, and the AND rule still rejected the tile.

#	Timestamp	Truth	Classical (conf)	RF (conf)	Final	Align dx, dy, deg
0	09:00:03.290	scratch	scratch (1.0)	ok (0.365)	FAIL	1.0, 2.2, -1.38
1	09:00:05.800	ok	ok (0.98)	ok (0.525)	PASS	3.22, 0.4, 0.8
2	09:00:08.250	ok	ok (0.947)	ok (0.37)	PASS	-0.39, 4.66, -1.59
3	09:00:11.300	missing_hole	missing_hole (0.848)	missing_hole (0.91)	FAIL	3.03, 1.36, 1.38
4	09:00:14.400	chipped_edge	chipped_edge (0.707)	chipped_edge (0.57)	FAIL	-5.18, 2.04, -2.84
5	09:00:17.500	colour_dev	colour_dev (1.0)	colour_dev (0.985)	FAIL	-1.39, 1.22, -0.93
6	09:00:20.000	ok	ok (0.983)	ok (0.43)	PASS	2.97, -2.64, 2.73
7	09:00:23.050	colour_dev	colour_dev (1.0)	colour_dev (0.93)	FAIL	2.41, -4.96, 2.16
8	09:00:25.550	ok	ok (0.984)	ok (0.455)	PASS	-1.48, 1.21, 0.89
9	09:00:28.600	missing_hole	missing_hole (0.858)	missing_hole (0.885)	FAIL	-0.8, -0.44, -1.02

8. Declared substitutions / not run here

■ **RoboDK:** external_apis/robodk_inspection.py drives a RoboDK station (targets InspectPose/RejectPose/HomePose, conveyor axis, Cam2D snapshot) with the same arm_inspect decision code. The RoboDK API is not installed here; the script was only import-checked and prints a notice when robodk is missing.

■ **CoppeliaSim (V-REP):** external_apis/coppelasim_remote_stub.py uses the ZeroMQ remote API to stream the trapezoidal joint targets and read the vision sensor. Not run here.

■ **ROS 2 Humble:** ros2_ws/src/arm_inspection/ (package.xml, setup.py, setup.cfg, resource marker, launch file, node.py publishing trajectory_msgs/JointTrajectory and a JSON result string). Syntax-checked with py_compile only.

9. Learnings / next steps

The rule-based pipeline is only as good as its registration: until I looked at blob statistics per class I was chasing false alarms that were really ECC residue on the hole ring. Measuring, then thresholding, beat guessing. Next: add a rotation-invariant scratch detector (ridge filter), grow the training set and try a small CNN, and replace the belt-stop cycle by tracking the item during motion so the arm cycle overlaps belt travel.

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